# Severity Prediction in Mental Health: LLM-based Creation, Analysis, Evaluation of a Novel Multilingual Dataset

Anonymous ACL submission

## Abstract

Large language models (LLMs) are increasingly employed across medical domains and lately even in mental health support platforms. However, relatively less research has been conducted on evaluating the use of LLMs in mental health support systems and in languages other than English. To fill this gap, we introduce a new multilingual dataset on mental health severity detection, evaluate and analyze it across seven languages (Greek, Turkish, French, Portuguese, German, Finnish, English). The findings indicate that LLMs exhibit varying performance levels in detecting mental health conditions and their severity across languages (e.g., best for Greek and worst for Turkish). Based on our error analyses, there is a need for further refinement and culturally nuanced training, and their use in mental health support platforms requires utmost care since their performance varies across languages. In addition, their use should always be supervised by human experts to prevent misdiagnosis during decision-making.

## 1 Introduction

Large language models (LLMs) such as GPT-3, BERT, and their multilingual variants are trained on vast amounts of text data and demonstrate an impressive ability to generate human-like text, perform complex language tasks, and exhibit a degree of contextual understanding previously unseen in AI systems (Devlin et al., 2019; Brown et al., 2020). One of the most promising applications of LLMs is in the healthcare domain, where they have the potential to revolutionize the detection, diagnosis, and treatment of health conditions and potentially assist the medical specialists with diagnosis.

Mental health issues (e.g., depression, anxiety, and post-traumatic stress disorder (PTSD)) are prevalent worldwide and pose significant challenges to public health (World Health Organization, 2021). Traditional methods for diagnosing mental health conditions often rely on self-reported surveys, clinical interviews, and standardized assessments conducted by trained professionals (Kessler and Üstün, 2004). While these methods are effective, they are also resource-intensive, time-consuming, and may not always be accessible to individuals in need, particularly for the speakers of languages beyond English.

In this context, the application of LLMs to detect mental health symptoms from textual data offers a compelling alternative. These models can analyze large volumes of text data (e.g., social media posts, forum discussions, and personal narratives) quickly to identify linguistic markers associated with mental health conditions (Guntuku et al., 2019; Chancellor et al., 2019). This capability opens up new avenues for early detection and intervention, providing valuable support to mental health professionals and potentially reaching out to the patients whose symptoms may be overlooked and/or save time (e.g., long waiting times).

Despite the potential benefits, the performance of LLMs in multilingual mental health symptom detection remains underexplored. Previous studies have primarily focused on English-language datasets, leaving a gap in our understanding of how these models perform in other linguistic contexts (Raihan et al., 2024). Our research questions are as follows:

- Can an LLM predict the severity mental health conditions in user-generated posts (English) accurately?
- When we automatically translate the mental health data set (English) into other languages with LLMs, can we expect a similar performance for predicting the severity levels of mental health conditions in other languages?

To address these questions, we assess a state-of-the-art multilingual LLM when predicting the

severity of mental health in English user-generated posts. Then, we automatically translate these posts from English into six languages, and re-assess the LLMs performance. Our research not only contributes to the development of more robust and inclusive AI-driven mental health diagnostic tools but also emphasizes the importance of culturally and linguistically sensitive approaches in mental health care beyond English. In sum, our contributions are:

- We develop a novel multilingual (English, Turkish, French, Portuguese, German, Greek, and Finnish) social media dataset (automatically translated from English by an LLM).[1]
- We conduct a comprehensive evaluation of the predictive power of an LLM detecting the severity of mental health conditions across the aforementioned languages.
- Finally, we present a detailed analysis of how diverse languages perform for the specific task and across data sets.

## 2 Related work

Early research in using NLP for mental health focused on detecting specific symptoms from textual data utilizing machine learning techniques to analyze social media posts, online health forum discussions, and other textual sources to identify markers of mental health conditions (e.g., depression, anxiety). (Guntuku et al., 2019). These approaches often rely on traditional machine learning algorithms and manually crafted features, lack the depth and nuance of modern LLMs.

On the other hand, LLMs have remarkable accuracy in detecting mental health symptoms by leveraging their ability to understand context and semantics at a deeper level. Examples include BioBERT (Lee et al., 2020), and ClinicalBERT (Huang et al., 2019), which are pre-trained on biomedical corpora or clinical notes. In contrast, models like MentalBERT (Ji et al., 2022), DisorBERT (Aragon et al., 2023), and SuicidalBERT (Anonymous) are pre-trained on mental health-related social media data. Additionally, research by Benton et al. (2017) showed that NLP can effectively assess depression and PTSD from clinical notes, further validating the utility of these models in a healthcare setting.

**Multilingual Capabilities of LLMs** Although the details about training and evaluation are not always transparent, the multilingual capabilities of LLMs enable these models to understand and generate text in various languages. A recent example is the XLM-R model, which has been trained on a vast amount of multilingual data and shows strong performance across multiple languages. According to Conneau et al. (2020), their model XLM-R outperforms previous models on a wide range of tasks, demonstrating that leveraging large-scale multilingual data can lead to improvements in cross-lingual understanding.

There are also examples of LLMs excelling in practical applications such as translation and multilingual question answering. For instance, a model created by Xue et al. (2021), is pre-trained on a diverse set of languages and demonstrates strong capabilities in translating and answering questions in multiple languages. The model's architecture and training methods allow it to generalize across languages effectively, providing robust performance even in languages with limited training data.

Despite these advancements, significant challenges remain in achieving truly equitable performance across all languages and handling culturally specific contexts accurately (Zhang et al., 2020). Languages with limited digital text data still pose a considerable challenge for LLMs, often resulting in lower performance and less reliable outputs. Addressing this issue requires more inclusive data collection practices and further research into transfer learning techniques that can better utilize limited resources (Doğruöz and Sitaram, 2022). Additionally, capturing cultural nuances and context-specific meanings is a complex task, as language is deeply intertwined with cultural and societal norms. Efforts to improve these aspects include developing more sophisticated algorithms and incorporating diverse and representative datasets (Doğruöz et al., 2023), ensuring that the benefits of multilingual LLMs are accessible to a broader range of users globally.

**LLMs for Mental Health** Using large language models (LLMs) for detecting mental health issues through social network posts represents a promising frontier in both artificial intelligence and mental health care. LLMs (e.g., GPT-4 and BERT) can analyze vast amounts of textual data from social media platforms automatically to identify linguistic patterns associated with mental health conditions such as depression, anxiety, and suicidal ideation. For instance, research by Shing et al. (2018) demon-

[1]The dataset will be publicly available upon acceptance.

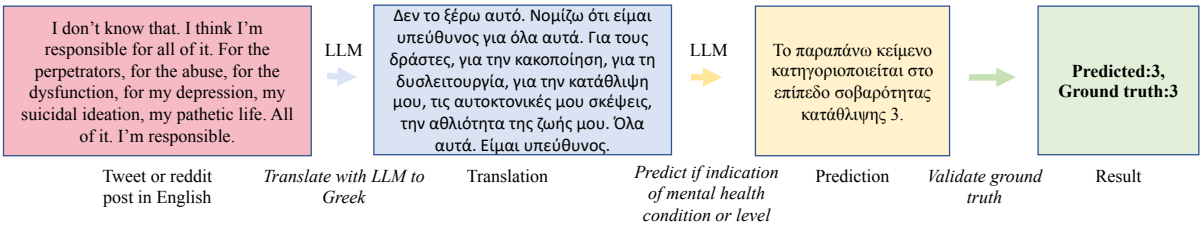



Figure 1: An illustration of our proposed methodology.

strates how machine learning models can assess suicide risk by analyzing language use in social media posts. The study shows that linguistic markers (e.g., increased use of first-person pronouns and negative emotion words) are predictive of higher suicide risk.

More recently, a plethora of social network datasets targeting mental health, have been available (Raihan et al., 2024). The authors gathered social media posts from Reddit and Twitter regarding depression, PTSD, schizophrenia, and eating disorders. Moreover, multiple models were fine-tuned on small-sized publicly available annotated mental health datasets by the authors to use them for labeling their introduced MentalHelp dataset. Nevertheless, the dataset includes only posts in English, and thus its use is restrictive, disallowing further research for multilingual scenarios.

## 3 Proposed Methodology

We present a comprehensive methodology for leveraging LLMs and social media posts (in English), in order to translate them into multiple languages and predict mental health conditions accordingly.

We first translate the social media posts across the proposed languages via an LLM. Then, these translations are fed into a prompt that asks the LLM to predict specific severity levels of mental health conditions. Finally, the predicted classes are compared with the ground truth labels and the evaluation metrics are computed.

Our goal is to propose a methodology that can be used to evaluate LLMs on multilingual detection of mental health conditions in languages where minimal relative linguistic resources are publicly available. An illustration of the proposed approach for evaluating LLMs for multilingual detection of mental health conditions is shown in Figure 1.

Next, we present the conducted experiments, including the used datasets, setup, results, and further analysis.

## 4 Datasets

We use the following datasets:

**DEP-RED** This dataset (Naseem et al., 2022) consists of posts from the social media platform Reddit, regarding different levels of depression. The posts (in English) are already labeled in terms of four levels of severity: minimal, mild, moderate, and severe depression. The majority of posts belong to the minimal severity level making it a highly imbalanced dataset.

**SUI-TWI** This dataset (Mirza Ibtihaj et al., 2020) contains texts from Twitter (in English), including 1,300 suicidal ideation tweets. The authors gathered posts that contained the word 'suicide' and labeled them manually into three classes "not suicidal", "indicative of suicidal ideation", and "indicative of a potential suicide attempt". In this dataset, the classes are balanced in terms of frequency of occurence.

We specifically selected multi-class datasets to make the task more challenging for the LLM, as binary problems would have been easier to answer. Table 1 presents some summarized statistics of the two datasets used in our research.

## 5 Experimental Setup

We use GPT3.5-turbo (Brown et al., 2020) and its API to translate the posts and predict the labels. The temperature parameter is set to 0, so the outcome is reproducible, regarding translations and predictions. The task is solved as a classification problem, where we compare the predicted classes with the ground-truth ones, reporting accuracy and f1-score. We experiment with seven languages from the following major language families: Turkish (Altaic), French, Portuguese (Romance), English, German (Germanic), Finnish (Uralic), Greek (Hellenic).

Table 2 shows the used prompts for predicting the severity levels.

| Dataset | Reference | Category | #Classes | Subset | #Instances | Labels (#Support) |
|---|---|---|---|---|---|---|
| DEP-RED | Naseem et al. (2022) | Depression | 4 | Test | 3553 | Minimum (2587)<br>Mild (290)<br>Moderate (394)<br>Severe (282) |
| SUI-TWI | Mirza Ibtihaj et al. (2020) | Suicide | 3 | Full | 1300 | Not Suicidal (515)<br>Suicidal Ideation (455)<br>Potential Suicide Attempt (330) |

Table 1: Benchmark datasets used in our study.

| Dataset | Prediction prompt |
|---|---|
| DEP-RED | "Categorize the following text with 1 of the 4 depression severity levels (0: Minimum, 1: Mild, 2: Moderate, 3: Severe)" |
| SUI-TWI | "Categorize to 0 if the text is not suicidal, to 1 if the text is almost related to suicidal ideation, and to 2 if the text indicates a potentially suicidal user" |

Table 2: The used prediction prompts.

### 5.1 Preliminary Prompting

Before exposing our LLM to any posts, either for translation or classification, we asked how it would classify them into different depression severity levels. The response was that, it would initially try to identify language patterns associated with depression, such as:

- Persistent negative emotions, such as sadness, or hopelessness.
- Self-criticism or feelings of worthlessness.
- Expressions of loneliness or social withdrawal.
- Changes in behavior or routines, as in sleep patterns or appetite.
- References to emotional pain or distress.

More specifically, LLM would try to adapt the four depression severity levels to fit the context of social media posts. For example:

- Level 1 (Minimum): Posts with minimal or occasional expressions of sadness or frustration.
- Level 2 (Mild): Posts indicating frequent negative emotions or noticeable changes in behavior.
- Level 3 (Moderate): Posts suggesting significant impairment in daily functioning or clear signs of distress.
- Level 4 (Severe): Posts indicating severe emotional distress, potential risk factors for self-harm, or complete social withdrawal.

We can infer that the LLM expects posts with very generic indications of negative signs. Similarly, for the second dataset, we expect the LLM to detect various linguistic and contextual features in the social media posts to classify them. Some key aspects that it considers per class are:

1. **Not Suicidal**:
   - General Tone: Neutral, positive, or typical negative expressions not related to self-harm or life-threatening situations.
   - Content: Discussions about daily activities, hobbies, interests, work, or social interactions without any reference to self-harm or death.
   - Language: Absence of words or phrases related to self-harm, death, hopelessness, or helplessness.
2. **Indicative of Suicidal Ideation**:
   - Expression of Hopelessness: Phrases indicating a lack of hope for the future, such as "I can't see any way out", "Things will never get better", or "There's no point in trying".
   - Self-Hatred or Worthlessness: Statements like "I'm a burden", "Everyone would be better off without me," or "I hate myself".
   - Mentions of Death or Dying: Expressions about wanting to die or not wanting to live, such as "I wish I were dead", "I want to end it all", or "Life isn't worth living".
   - Social Withdrawal: Indications of isolating oneself from others, such as "I don't want to see anyone", or "I prefer to be alone all the time".

| | English | | Turkish | | French | | Portuguese | | German | | Greek | | Finnish | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Class** | **Acc** | **F1** | **Acc** | **F1** | **Acc** | **F1** | **Acc** | **F1** | **Acc** | **F1** | **Acc** | **F1** | **Acc** | **F1** |
| MINIMUM | 0.98 | 0.25 | 0.93 | 0.05 | **0.99** | 0.27 | 0.98 | 0.20 | 0.97 | **0.38** | 0.99 | 0.14 | 0.98 | 0.15 |
| MILD | 0.04 | 0.07 | 0.02 | 0.01 | 0.05 | 0.08 | 0.04 | 0.05 | **0.08** | **0.12** | 0.04 | 0.06 | 0.05 | 0.08 |
| MODERATE | 0.13 | 0.17 | 0.10 | 0.17 | 0.15 | 0.21 | 0.12 | 0.17 | **0.16** | **0.23** | 0.14 | 0.23 | 0.12 | 0.17 |
| SEVERE | 0.13 | 0.22 | 0.13 | 0.19 | 0.15 | 0.23 | 0.12 | 0.20 | **0.16** | **0.22** | 0.16 | 0.20 | 0.12 | 0.20 |
| Average | 0.19 | 0.17 | 0.12 | 0.11 | 0.21 | 0.20 | 0.18 | 0.16 | **0.27** | **0.24** | 0.15 | 0.16 | 0.15 | 0.15 |

Table 3: Accuracy and F1-Score for each class and language on the DEP-RED dataset (depression). Last row shows average accuracy and weighted average F1-score. The best per measure is shown in bold.

| | English | | Turkish | | French | | Portuguese | | German | | Greek | | Finnish | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Class** | **Acc** | **F1** | **Acc** | **F1** | **Acc** | **F1** | **Acc** | **F1** | **Acc** | **F1** | **Acc** | **F1** | **Acc** | **F1** |
| NOT SUICIDAL | **0.78** | **0.63** | 0.58 | 0.50 | 0.58 | 0.62 | 0.63 | 0.58 | 0.56 | 0.61 | 0.58 | **0.63** | 0.57 | 0.55 |
| NEARLY RELATED | 0.46 | **0.58** | 0.36 | 0.47 | 0.43 | 0.42 | 0.43 | 0.49 | 0.39 | 0.41 | **0.49** | 0.49 | 0.35 | 0.38 |
| POTENTIAL | **0.69** | 0.47 | 0.54 | 0.07 | 0.54 | 0.50 | 0.54 | 0.47 | 0.51 | 0.39 | 0.65 | **0.55** | 0.49 | 0.43 |
| Average | **0.57** | **0.57** | 0.43 | 0.38 | 0.52 | 0.52 | 0.52 | 0.52 | 0.49 | 0.48 | 0.56 | 0.56 | 0.46 | 0.46 |

Table 4: Accuracy and F1-Score for each class and language on the SUI-TWI dataset (suicide). Last row shows average accuracy and weighted average F1-score. The best per measure is shown in bold.

3. **Indicative of a Potential Suicide Attempt**:

   - Detailed Plans or Intentions: Descriptions of specific methods or plans to commit suicide, such as “I have pills ready," “I’m going to jump off a bridge," or “I’ve written my goodbye letter."
   - Goodbye Messages: Posts that read like final farewells, such as “Goodbye everyone, it’s been a pleasure," or “I’m sorry for everything, I won’t be around much longer."
   - Previous Attempts: References to past suicide attempts, particularly if combined with current suicidal thoughts, such as “I’ve tried before and failed, but not this time."
   - Lethal Means: Mentions of acquiring or possessing means to commit suicide, like “I bought a gun," “I have enough pills," or “I found a rope."

When asked, the LLM presents a detailed description of the three classes, which seems promising.

## 5.2 Experimental results

The results are organized and presented in the following subsections: classification in the source languages, classification in the target language, classification across languages, and further analysis of LLMs performance with multilingual examples.

### 5.2.1 Classification in the source language

Initially, we experimented with the data in their source language (English), to set the baseline performance. That is, no translation step has been performed at this stage. The first column in Tables 3 and 4 presents the results on depression and suicide detection tasks respectively. The best F1 in both tasks is achieved for the lowest severity/indication. In DEP-RED, the best performance is achieved for the two edges of lowest (F1=0.25) and highest severity (F1=0.22). In SUI-TWI, on the other hand, F1 drops as the suicidal likelihood increases, from 0.63 to 0.47. Overall, the performance is significantly better for the task of suicide ideation detection (F1=0.57) compared to the depression task (F1=0.17). This can be attributed to the fact that detecting specific levels of depression is harder than in the case of suicidal ideation, as depression severity levels are considered less distinct. Therefore, it is probably more challenging for the LLM to distinguish.

### 5.2.2 Classification in the target language

The worst results in both datasets are achieved when classification is performed on Turkish LLM translations. This might be explained by the limited number of Turkish resources in health, especially in the domain of mental health (Çöltekin et al., 2023). Mental health terminologies and nuances may not be well-represented in the available Turk-

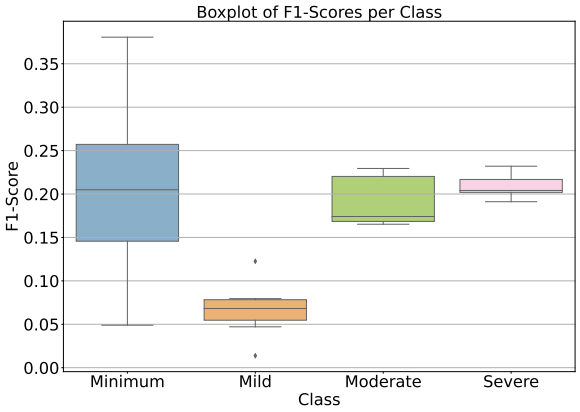



Figure 2: F1-score for DEP-RED on all languages.

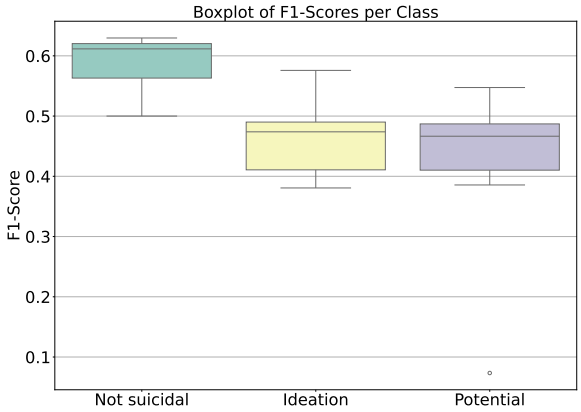



Figure 3: F1-score for SUI-TWI on all languages.

ish corpora, making it difficult for LLMs to grasp the context accurately. On the other end, the best results were not always obtained for English (the source language)(§5.2.1) either. In DEP-RED, the best F1 was established for German and the best accuracy was for French. In SUI-TWI, the best results were shared between English and Greek, with English being better in the "Suicidal Ideation" category and the Greek target language being better in the "Potential Suicidal Attempt" category (similar performance in the "Not suicidal" category).

For languages like German and French, the availability and quality of training data in the mental health domain might be higher on the internet, these languages may have better-annotated datasets, more comprehensive lexicons, and richer semantic resources. In addition, some languages might inherently structure information in ways that make them more amenable to classification tasks (Gasparetto et al., 2022). For instance, German and French may express suicidal ideation and attempts in ways that are more straightforward or unambiguous, aiding in better classification.

#### 5.2.3 Classification across languages

To have a better picture of the performance across languages than a single average, Figures 2 and 3 show the boxplots of F1 across languages for both tasks. In DEP-RED, the lowest (minimum) severity is described by a wide range from 0.05 (Turkish) to 0.38 (German). Mild severity is mishandled across languages while moderate and severe achieve a relatively higher score. We point out that DEP-RED is largely imbalanced, making it harder to predict correctly.

In SUI-TWI, we observe that the range of performance across languages is similarly limited across classes. The best performance is achieved for the "non-suicidal" category while slightly worse performance is achieved for the other two classes. Again, the majority of tweets belong to the first class with no suicidal indication.

#### 5.2.4 Further analysis with examples

Figure 4 presents two instances of the depression dataset and their corresponding translations in Greek. We chose Greek as one of the languages observing the best performance and marked words and their translations with similar colors for better visibility. First, both translations appear to be accurate and convey the same meaning as the original Greek text. There are no significant differences that would alter the understanding of the texts. The first segment contains explicit mentions of severe depression symptoms such as "want to give up" and "feeling of wanting to die." These statements clearly indicate a severe level of depression, which is why both the ground truth and prediction were classified as severe.

The second segment describes physical sensations, panic, and anxiety but does not express a severe depressive state. The ground truth classified this as minimal depression, likely because the primary issues are related to panic and anxiety rather than depression. The LLM predicted a moderate level of depression for the second segment, possibly because it picked up on the words "panicked" and "anxiety", which are associated with higher levels of distress. However, these symptoms are more indicative of anxiety disorders rather than depression. The discrepancy in the second prediction can be attributed to the LLM's interpretation of anxiety and panic as indicative of moderate depression, whereas the ground truth assessment con-

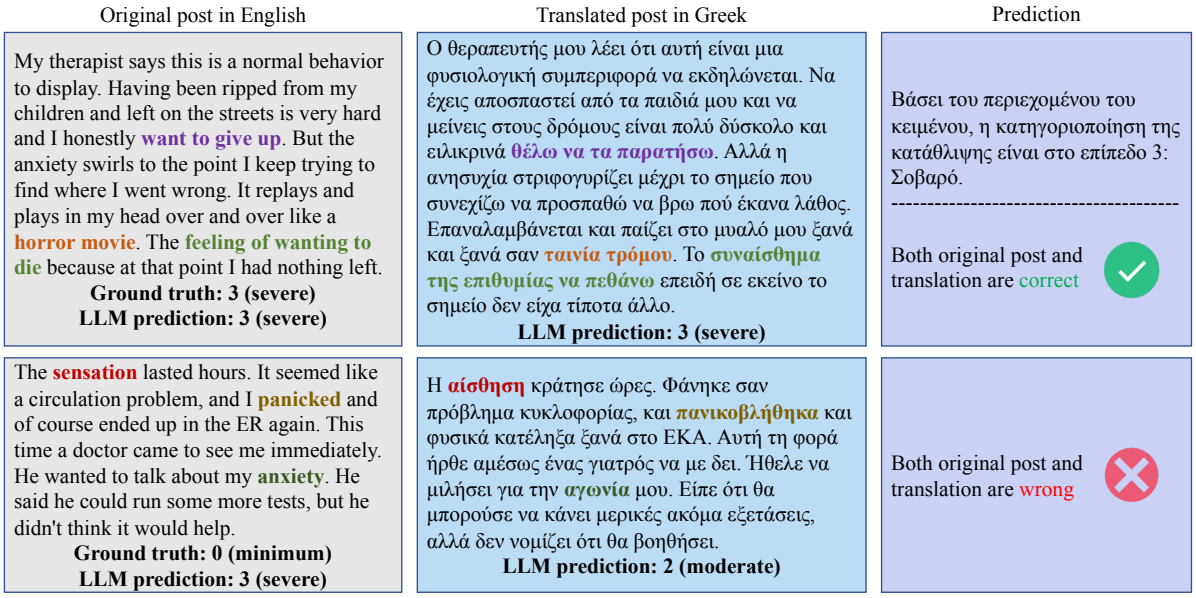



Figure 4: An example of translation (English to Greek) from the depression dataset. The original words and their translations are colored similarly.

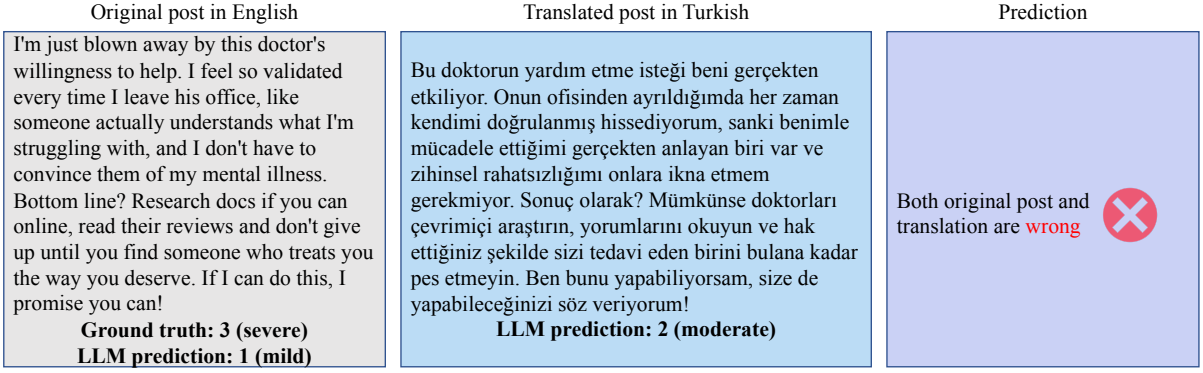



Figure 5: An example of translation (English to Turkish) from the depression dataset.

siders these symptoms in the context of a panic or anxiety disorder with minimal depression. This highlights the challenge of distinguishing between different types of mental health issues based solely on short textual descriptions.

Figure 5 presents an example of translation in Turkish. In this example, the original post in English was labeled as "severe" from the ground truth data, but the LLM predicted the severity level as "mild" for English and "moderate" for Turkish. A closer look at the example reveals that the original label that came with the dataset was probably not correctly assigned. In this case, LLM corrected this error by lowering the level of severity for both English and Turkish (although the levels of severity were slightly different for each language). However, since we compare the predictions with ground truth we show that both the original post and translation failed to predict correctly the severity level.

Lastly, Figure 6 presents two instances of the suicide dataset. Regarding the first example, while the translation from English to Greek appears to be correct in terms of conveying the same message, the English text is classified as "Potential suicide attempt", while the Greek text is classified as "Suicidal ideation". The LLM might be interpreting the context differently due to nuances in the languages. In English, "attempted suicide" explicitly suggests a recent suicide attempt, which clearly falls under "Potential suicide attempt." In Greek, the model might be weighing other parts of the text more heavily, leading to a classification of "Suicidal ideation." This could be due to differences in the training data, language model intricacies, or contextual emphasis.

For the second example, we observe that the

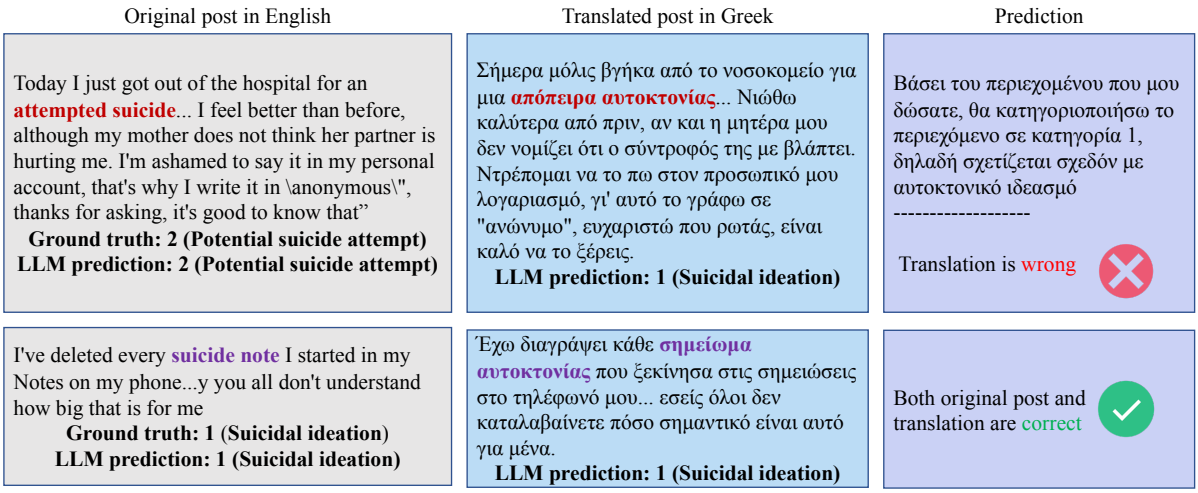



Figure 6: An example of translation (English to Greek) from the suicide dataset. The original words and their translations are colored similarly.

translation is accurate and conveys the same message as the original English text. Both the English and Greek texts are classified consistently and correctly as "Suicidal ideation".

### 5.3 Cost of experiments

The total cost of credits using GPT3.5-turbo via its API was less than $30 which shows that minimal resources were required to conduct our experiments, without the need for expensive GPU infrastructure or fine-tuning. Our methodology for utilizing resources is potentially promising for extending medical data sets in English into other languages as well.

## 6 Conclusion and Future Work

In our study, we focused on the ability of an LLM to predict the severity of the mental health conditions in user-generated posts, in English (source) and in six target languages (machine translated with an LLM). To address this question, we translated posts from English to other languages, and we prompted the LLM to predict the posts' class per language for two tasks related to mental health conditions. Our findings show that the LLM performance ranges depending on the language. Also, the best performance is not necessarily achieved in the source (English) language. When predicting depression severity, best performance was in German. For predicting the indications of suicide in tweets, best performance was in English and Greek. The worst performance in both tasks was achieved in Turkish.

Our experiments also question the quality of ground truth labels that came along with the data sets. This is shown in Figure 5 for Turkish, where the plausible LLM prediction differs from the ground truth.

Next steps in this direction comprise the application of this methodology to a broader range of languages, (including a variety of low-resource languages) and new data sets from other social media platforms. Additionally, multilingual LLMs, (potentially fine-tuned on mental health data) could further enhance the practical utility of this approach.

Considering the variation in the performance of the LLM across languages, there is a need for utmost precautions not to rely on LLMs solely in any health care setting. As stated by Stade et al. (2024), diagnosis of mental health should never be left to automatic systems only and should never replace the diagnosis by human professionals due to possible errors and/or misdiagnosis illustrated above. Despite the hype of using LLMs in the mental healthcare domain for commercial applications, there is an increasing need for more research to understand the pitfalls and dangers of using LLMs across languages, diseases, and patient populations for mental health care. Before there are substantial research findings approved by experts of mental health, it is probably wise not to introduce LLMs as alternatives to replace patient-doctor sessions for diagnosing or treating mental care issues across languages. However, our approach could be useful to train the health care professionals (especially in languages other than English) in the mental healthcare domain instead of assisting the patients directly.

## Limitations

**Translation** Creating a multilingual dataset requires significant resources. In this work, we used a popular LLM like GPT3.5 to translate posts. Translating using only an LLM and not having an expert or native-language human resources introduces a small loss of information that in some cases affects the final results.

**Evaluation** Automatically evaluating the performance of LLMs is by definition a hard task. In order to measure the performance we search for the label in the LLM output. Whenever no label is detected we count it as the minimum label (class: 0) for the depression dataset and not suicidal (class: 0) for the suicide dataset.

**Potential risks** The quality of publicly available datasets, especially in sensitive areas like the mental health care domain is of great importance for prediction tasks. The data sets we employed as a basis in our study, along with our created multilingual data should be used with utmost care and only for assisting the health care specialists instead of diagnosing patients directly.

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